FINITE DIMENSIONALITY OF TANGENT AND INFINITESIMAL AUTOMORPHISM SPACES OF THE COMPLEXES STACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5765. Standing situation.
- Section artin-section-tangent-spaces (canonical tag 07WY). cited at statement line 5772. Definitions of tangent space and Inf.
- Lemma lemma-complexes-RS-star (canonical tag 0DLK). cited at proof lines 5777-5778. (RS*) of the complexes stack.
- Section artin-section-RS-star (canonical tag 0CXN). cited at proof line 5779. Definition of (RS*).
- Lemma artin-lemma-deformation-category (canonical tag 07WU). cited at proof lines 5782-5783. (RS) makes the fibres deformation categories.
- Remark remark-complexes-base-change (canonical tag 0DLI). cited at proof line 5790. X_0 as 2-fibre product.
- Lemma artin-lemma-fibre-product-tangent-spaces (canonical tag 07X2). cited at proof lines 5793-5794. Tangent/Inf of 2-fibre products.
- Lemma artin-lemma-finite-dimension (canonical tag 07X1). cited at proof lines 5797-5798. Finite dimensionality for B and Spec(k).
- Section spaces-more-morphisms-section-thickenings (canonical tag 05ZJ). cited at proof lines 5813-5814. Equivalent small etale topoi under first order thickenings.
- Lemma spaces-more-morphisms-lemma-thickening-relatively-perfect (canonical tag 0DL4). cited at proof lines 5815-5816. Tangent space = lift isomorphism classes.
- Lemma defos-lemma-first-order-defos-complex (canonical tag 0DIZ). cited at proof lines 5819 and 5827. Ext^1 computes the tangent space; Ext^0 surjects onto Inf.
- Lemma spaces-perfect-lemma-identify-pseudo-coherent-noetherian (canonical tag 08IK). cited at proof lines 5836-5837. E_0 in D^-_Coh on the Noetherian X_0.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Lemma spaces-perfect-lemma-ext-finite (canonical tag 0D0T). cited at proof lines 5841-5842. Ext groups finite dimensional.

Source reference. Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5765: the lemma is phrased in this situation; load-bearing.

Source reference. Original section 07WY: sources/stacks-chow/tags.txt, source lines 10058–10058. Cited at statement line 5772: the objects whose finite dimensionality is claimed; load-bearing.

Source reference. Original lemma 0DLK: sources/stacks-chow/tags.txt, source lines 17050–17050. Cited at proof lines 5777-5778: X satisfies (RS*), hence (RS); load-bearing.

Source reference. Original section 0CXN: sources/stacks-chow/tags.txt, source lines 16213–16213. Cited at proof line 5779: definition of the property; load-bearing.

Source reference. Original lemma 07WU: sources/stacks-chow/tags.txt, source lines 10054–10054. Cited at proof lines 5782-5783: the predeformation categories are deformation categories so tangent/Inf make sense; load-bearing.

Source reference. Original remark 0DLI: sources/stacks-chow/tags.txt, source lines 17048–17048. Cited at proof line 5790: X_0 is the 2-fibre product with Spec(k) over B; load-bearing for the reduction.

Source reference. Original lemma 07X2: sources/stacks-chow/tags.txt, source lines 10062–10062. Cited at proof lines 5793-5794: reduces to the three factors; load-bearing.

Source reference. Original lemma 07X1: sources/stacks-chow/tags.txt, source lines 10061–10061. Cited at proof lines 5797-5798: B and Spec(k) have finite dimensional tangent spaces; load-bearing.

Source reference. Original section 05ZJ: sources/stacks-chow/tags.txt, source lines 7693–7693. Cited at proof lines 5813-5814: X_0 and X_epsilon have canonically equivalent small etale topoi; load-bearing.

Source reference. Original lemma 0DL4: sources/stacks-chow/tags.txt, source lines 17034–17034. Cited at proof lines 5815-5816: the tangent space is the set of lift isomorphism classes; load-bearing.

Source reference. Original lemma 0DIZ: sources/stacks-chow/tags.txt, source lines 16959–16959. Cited at proof lines 5819 and 5827 (twice): identifies T F with Ext^1 and surjects Ext^0 onto Inf; load-bearing twice.

Source reference. Original lemma 08IK: sources/stacks-chow/tags.txt, source lines 10818–10818. Cited at proof lines 5836-5837: E_0 lies in D^-_Coh; load-bearing.

Source reference. Original lemma 0D0T: sources/stacks-chow/tags.txt, source lines 16323–16323. Cited at proof lines 5841-5842: Ext^0 and Ext^1 are finite dimensional; load-bearing.

Lemma 0.2. *In Situation 0DLE assume that S is a locally Noetherian scheme and $B \to S$ is locally of finite presentation. Let k be a finite type field over S and let $x_0 = (\mathrm{Spec}(k), g_0, E_0)$ be an object of $\mathcal{X} = \mathit{Complexes}_{X/B}$ over k. Then the spaces $T\mathcal{F}_{\mathcal{X},k,x_0}$ and $\mathit{Inf}(\mathcal{F}_{\mathcal{X},k,x_0})$ (Artin's Axioms, Section 07WY) are finite dimensional.*

Proof. Observe that by Lemma 0DLK our stack in groupoids $\mathcal{X}$ satisfies property (RS*) defined in Artin's Axioms, Section 0CXN. In particular $\mathcal{X}$ satisfies (RS). Hence all associated predeformation categories are deformation categories (Artin's Axioms, Lemma 07WU) and the statement makes sense.

In this paragraph we show that we can reduce to the case $B = \mathrm{Spec}(k)$. Set $X_0 = \mathrm{Spec}(k) \times_{g_0,B} X$ and denote $\mathcal{X}_0 = \mathit{Complexes}_{X_0/k}$. In Remark 0DLI we have seen that $\mathcal{X}_0$ is the 2-fibre product of $\mathcal{X}$ with $\mathrm{Spec}(k)$ over B as categories fibred in groupoids over $(\mathit{Sch}/S)_{fppf}$. Thus by Artin's Axioms, Lemma 07X2 we reduce to proving that B, $\mathrm{Spec}(k)$, and $\mathcal{X}_0$ have finite dimensional tangent spaces and infinitesimal automorphism spaces. The tangent space of B and $\mathrm{Spec}(k)$ are finite dimensional by Artin's Axioms, Lemma 07X1 and of course these have vanishing Inf. Thus it suffices to deal with $\mathcal{X}_0$.

Let $k[\epsilon]$ be the dual numbers over k. Let $\mathrm{Spec}(k[\epsilon]) \to B$ be the composition of $g_0 : \mathrm{Spec}(k) \to B$ and the morphism $\mathrm{Spec}(k[\epsilon]) \to \mathrm{Spec}(k)$ coming from the inclusion $k \to k[\epsilon]$. Set $X_0 = \mathrm{Spec}(k) \times_B X$ and $X_\epsilon = \mathrm{Spec}(k[\epsilon]) \times_B X$. Observe that X_ϵ is a first order thickening of X_0 flat over the first order thickening $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\epsilon])$. Observe that X_0 and X_ϵ give rise to canonically equivalent small étale topoi, see More on Morphisms of Spaces, Section 05ZJ. By More on Morphisms of Spaces, Lemma 0DL4 we see that $T\mathcal{F}_{\mathcal{X}_0,k,x_0}$ is the set of isomorphism classes of lifts of E_0 to X_ϵ in the sense of Deformation Theory, Lemma 0DIZ. We conclude that

$$T\mathcal{F}_{\mathcal{X}_0,k,x_0} = \mathrm{Ext}^1_{\mathcal{O}_{X_0}}(E_0, E_0)$$

Here we have used the identification $\epsilon k[\epsilon] \cong k$ of $k[\epsilon]$-modules. Using Deformation Theory, Lemma 0DIZ once more we see that there is a surjection

$$\mathrm{Inf}(\mathcal{F}_{\mathcal{X},k,x_0}) \leftarrow \mathrm{Ext}^0_{\mathcal{O}_{X_0}}(E_0, E_0)$$

of k-vector spaces. As E_0 is pseudo-coherent it lies in $D^-_{Coh}(\mathcal{O}_{X_0})$ by Derived Categories of Spaces, Lemma 08IK. Since E_0 locally has finite tor dimension and X_0 is quasi-compact we see $E_0 \in D^b_{Coh}(\mathcal{O}_{X_0})$. Thus the Exts above are finite dimensional k-vector spaces by Derived Categories of Spaces, Lemma 0D0T. □